FoodRescue AI: Autonomous Surplus-Food Detection and Redistribution System

A Mini Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINNERING

Or

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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DECLARATION

We hereby declare that the mini-project report entitled **"FoodRescue AI: Autonomous Surplus-Food Detection and Redistribution System"** is a record of the work carried out by our team in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

The project focuses on developing a technology-based platform to manage surplus food donations, analyze food urgency, identify suitable recipient organizations, coordinate redistribution, and monitor delivery progress. The work presented in this report is based on our project implementation, testing, and analysis.

We declare that this report has been prepared for academic purposes and that all external resources used have been acknowledged in the References section.

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CERTIFICATE

This is to certify that the mini-project report entitled **"FoodRescue AI: Autonomous Surplus-Food Detection and Redistribution System"** is a Bonafide work done and submitted by **P Poojith Varma Sai – 2420030749, G. Prashanth Kumar – 2420090072, and V. Harshith – 2420030020**, in partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in the Department of Computer Science Engineering, K L (Deemed to be) University, during the academic year **2026–2027**.

Signature of the Course Co-ordinator **Signature of HOD**

ACKNOWLEDGEMENT

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1. ABSTRACT

FoodRescue AI: Autonomous Surplus-Food Detection and Redistribution System is a web-based application developed to reduce food wastage by connecting surplus-food providers with suitable recipient organizations. The system provides a centralized platform to register surplus food, analyze its urgency, identify potential recipients, coordinate dispatch, and monitor delivery progress.

The application is developed using React, TypeScript, and Vite for the frontend, FastAPI and Python for the backend, and SQLite for database management. Google Gemini AI is integrated to support food urgency and perishability analysis, while a rule-based fallback provides analysis when the external AI service is unavailable. A five-factor matching algorithm evaluates category compatibility, recipient need, capacity, distance, and availability to calculate recipient suitability scores.

The platform supports four user roles: Food Provider, Recipient, Logistics Coordinator, and Administrator. It includes role-based access control, redistribution management, pickup tracking, notifications, map visualization, and administrative statistics. The database maintains records of food submissions, recipient matches, redistributions, pickup tasks, and notifications.

According to the project's functional audit reports, the main workflows passed the reported end-to-end and regression tests. The current prototype uses simulated temperature and EV battery readings, while SMS and email services depend on external configuration. FoodRescue AI demonstrates how AI-assisted analysis and automated workflows can improve the organization and traceability of surplus-food redistribution.

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2. INTRODUCTION

Food wastage is a major concern in today's society, as large quantities of food remain unused in hotels, restaurants, and other food establishments while many people struggle to access sufficient food. One of the main reasons for this problem is the lack of an efficient system to connect surplus-food providers with organizations that can distribute food to people in need. Manual coordination can also cause delays, difficulties in identifying suitable recipients, and problems in tracking deliveries.

FoodRescue AI: Autonomous Surplus-Food Detection and Redistribution System is a web-based application developed to address these challenges. The project provides a centralized platform where food providers can register surplus food, analyze its urgency, identify suitable recipient organizations, and coordinate the redistribution process. It uses artificial intelligence to support food analysis and a five-factor matching algorithm to evaluate recipient suitability based on food category, recipient need, capacity, distance, and availability.

The application is developed using React and TypeScript for the frontend, FastAPI and Python for the backend, and SQLite for data storage. It also integrates Google Gemini AI for food analysis, with a rule-based fallback when the external AI service is unavailable. The system supports food providers, recipients, logistics coordinators, and administrators through role-based access and dedicated workflows. Features such as pickup tracking, notifications, map visualization, and administrative statistics help users monitor the redistribution process.

The main objective of FoodRescue AI is to make surplus-food redistribution more organized, efficient, and traceable through technology. This report explains the problem addressed by the project, reviews related systems, describes the requirements and implementation, presents the results of functional testing, and discusses the conclusion and future improvements.

3. LITERATURE SURVEY

1. FoodCloud

FoodCloud is a platform that connects businesses with surplus food to charities and community organizations. It helps ensure that usable surplus food reaches people who need it instead of being discarded. The platform supports coordination between food donors and recipient organizations. This project is related to FoodRescue AI because both systems focus on food donation, recipient coordination, and reducing food waste. FoodRescue AI additionally implements AI-assisted food analysis and a five-factor recipient matching algorithm.

Reference: https://www.food.cloud/

2. Too Good To Go

Too Good To Go is a food-saving platform that allows users to discover and purchase surplus food from participating food businesses. It helps businesses make use of unsold food and gives consumers access to food at reduced prices. The platform uses digital discovery and location-based services to connect users with available food. It demonstrates how technology can help reduce food waste. FoodRescue AI differs by focusing on food donations, recipient organizations, and logistics tracking rather than consumer purchases.

Reference: https://www.toogoodtogo.com/

3. OLIO

OLIO is a sharing platform that enables people and businesses to share surplus food and other reusable items with nearby community members. It encourages local participation and helps prevent usable products from being thrown away. Its location-based sharing approach helps connect people who have surplus items with those who can use them. This concept is relevant to FoodRescue AI because the project also uses location information to support food redistribution. FoodRescue AI adds role-based workflows and delivery lifecycle management.

Reference: https://olioapp.com/en/

4. Food Rescue US

Food Rescue US operates a web-based platform that connects food donors, volunteers, and social service agencies. It coordinates the movement of surplus food from businesses to organizations serving people experiencing food insecurity. The platform demonstrates the importance of scheduling, coordination, and delivery information in food recovery. FoodRescue AI follows a similar redistribution workflow and maintains records of pickup

and delivery status. It additionally includes AI-assisted urgency analysis and a five-factor recipient matching system.

Reference: https://foodrescue.us/

5. Feeding America

Feeding America operates a network of food banks, food pantries, and meal programs that distribute food to communities in need. Its network works with food businesses, manufacturers, retailers, and other partners to collect and distribute donated food. This model demonstrates the importance of coordination between donors and recipient organizations. FoodRescue AI applies a software-based approach to similar coordination activities by managing donations, recipient matching, and redistribution status. The two systems share the broader goal of improving access to food.

Reference: https://www.feedingamerica.org/our-work/food-bank-network

6. Copia

Copia is a food recovery platform focused on helping businesses manage surplus food and connect it with nonprofit organizations. Its approach highlights the role of technology in coordinating food donations and improving the visibility of recovery activities. Efficient food recovery requires information about available food, potential recipients, and collection arrangements. These requirements are closely related to FoodRescue AI's donation and logistics workflows. FoodRescue AI further demonstrates automated matching, AI-assisted analysis, and database-based lifecycle tracking.

Reference: https://www.gocopia.com/

7. K-12 Food Rescue

K-12 Food Rescue supports schools in redirecting eligible surplus food to people in need instead of sending it to landfills. It provides resources and tools that help schools organize food-sharing activities and measure their results. The platform demonstrates how organized food recovery can involve institutions and community organizations. This approach is relevant to FoodRescue AI because both encourage better management of surplus food and tracking of recovery activities. FoodRescue AI is designed as a broader web application for providers, recipients, logistics coordinators, and administrators.

Reference: https://www.foodrescue.net/

8. ReFED Insights Engine

ReFED provides data, research, and analytical resources related to food waste reduction. Its Insights Engine helps businesses, policymakers, and other stakeholders understand food waste and evaluate possible solutions. It emphasizes the importance of using reliable data to make decisions and assess the impact of interventions. This approach is relevant to FoodRescue AI's administrative dashboard and impact indicators. However, ReFED is primarily a data and solutions resource, whereas FoodRescue AI focuses on managing the operational workflow of surplus-food redistribution.

Reference: https://refed.org/food-waste/insights-engine

9. UNEP Food Waste Index Report 2024

The United Nations Environment Programme's Food Waste Index Report 2024 examines food waste across households, food services, and retail. It discusses methods for measuring food waste and highlights opportunities for reducing it through coordinated action. The report provides research context for developing digital systems that improve food recovery and responsible resource use. It is relevant to FoodRescue AI because the project aims to organize surplus-food redistribution and maintain records of donation activity. The report also emphasizes the importance of measuring impact using reliable data.

Reference: https://www.unep.org/resources/publication/food-waste-index-report-2024

Literature survey summary:

The reviewed platforms demonstrate different approaches to reducing food waste, including food donation networks, local food sharing, food recovery coordination, and data-driven analysis. They show that digital tools can help connect food providers with recipients and improve the organization of food recovery activities. Based on these approaches, FoodRescue AI combines surplus-food registration, AI-assisted urgency analysis, five-factor recipient matching, dispatch approval, pickup tracking, and administrative monitoring in a single application. Its main focus is the integration of these functions into a testable web-based prototype. Unlike established operational networks, the current implementation does not independently verify real-world food deliveries or environmental savings.

4. Problem Statement and Objectives

Problem Statement:

The lack of an efficient, intelligent system to identify surplus food and redistribute it to people in need leads to food waste and delays in food donation.

Problem Description:

Traditional food redistribution methods often face difficulties in assessing food urgency, matching suitable recipients, and coordinating pickups and deliveries. FoodRescue AI addresses these challenges through AI-based analysis and automated recipient matching.

Objectives:

1. Reduce Food Waste: Identify surplus food and support its timely redistribution to people in need.
2. Analyze Food Using AI: Assess food details, perishability, and urgency using AI, with a rule-based fallback.
3. Automate Recipient Matching: Match food donations with suitable recipients based on category compatibility, need, capacity, distance, and availability.
4. Improve Logistics Coordination: Support dispatch approval, pickup task generation, and delivery status tracking.
5. Track Redistribution Activities: Provide notifications, maps, and impact statistics to monitor food rescue operations.
6. Support Multiple Users: Provide separate access for food providers, recipients, logistics coordinators, and administrators.

5. HARDWARE AND SOFTWARE REQUIREMENTS

- **Hardware Requirements**

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB minimum, 8 GB recommended
Storage	10 GB of free disk space
System	Desktop computer or laptop
Internet	Stable internet connection
Display	Standard monitor or laptop display

- **Software Requirement**

Software / Technology	**Purpose**
Operating System	Windows 10/11, Linux, or macOS
Frontend	React, TypeScript, Vite, Tailwind CSS
Backend	Python, FastAPI
Database	SQLite
AI Integration	Google Gemini API
Code Editor	Visual Studio Code
Web Browser	Google Chrome or Microsoft Edge
Mapping	Leaflet and OpenStreetMap
Routing	OSRM with Haversine distance fallback
API Testing	Postman or Swagger UI
Version Control	Git and GitHub

6. IMPLEMENTATION

Frontend: The frontend of FoodRescue AI is developed using React, TypeScript, Vite, and Tailwind CSS. It provides interfaces for food providers, recipients, logistics coordinators, and administrators to manage food donations and redistribution.

Backend: The backend is developed using Python and FastAPI. It handles food submissions, authentication, AI-based food analysis, recipient matching, pickup management, and other application operations. SQLite is used to store application data.

API Integration: Google Gemini API supports AI-based food analysis. Leaflet and OpenStreetMap support map visualization, while OSRM provides route information. A Haversine-based fallback supports distance calculations when routing services are unavailable.

Frontend Files

- index.html – Main HTML entry point for the web application.
- src/ – Contains frontend source code and application components.
- package.json – Contains project dependencies and scripts.
- package-lock.json – Records the installed npm dependency versions.
- vite.config.ts – Configures the Vite development server and build process.
- tsconfig.json – Contains TypeScript compiler settings.
- tsconfig.app.json – Contains TypeScript settings for the application.
- tsconfig.node.json – Contains TypeScript settings for Node-related configuration.

Backend Files

- backend/app/main.py – Initializes the FastAPI application and registers API routers.
- backend/run.py – Starts the backend server.
- backend/requirements.txt – Lists the Python dependencies required by the backend.
- backend/foodrescue.db – SQLite database used to store application data.
- backend/app/core/ – Contains backend configuration, database connection, and security logic.
- backend/app/models/ – Contains application database models.
- backend/app/routers/ – Contains API route handlers.
- backend/app/schemas/ – Contains request and response validation schemas.
- backend/app/services/ – Contains backend service logic.

API Integration

- Gemini API: Used for AI-based food analysis, with a rule-based fallback when the external AI service is unavailable.
- Leaflet and OpenStreetMap: Used to display maps and locations.
- OSRM: Used to obtain route information for delivery planning.
- Haversine distance calculation: Used as a fallback for calculating approximate distances.
- Notification services: Support in-app notifications, with external SMS and email delivery dependent on configuration.

7. RESULTS & DISCUSSION

- The FoodRescue AI system was developed to support the identification and redistribution of surplus food through a web-based application. The frontend provides separate interfaces for food providers, recipients, logistics coordinators, and administrators.
- The system supports food submission, AI-based perishability analysis, recipient matching, redistribution approval, pickup tracking, and notifications. Map visualization helps users view locations and access route information.
- The recipient matching mechanism ranks potential recipients using multiple factors, including proximity, capacity, urgency, and compatibility. The system also maintains food redistribution records and provides impact statistics to monitor application activities.
- The backend uses FastAPI and SQLite to process requests and store application data. Automated tests reported by the project audit passed, while external SMS and email delivery depend on service credentials.
- Overall, FoodRescue AI provides an integrated approach to managing surplus food redistribution and improving coordination between food providers, recipients, and logistics teams.

8. CONCLUSION & FUTURE SCOPE

Conclusion:

FoodRescue AI was developed to reduce food waste by improving the identification, matching, and redistribution of surplus food. The system integrates React, TypeScript, Python FastAPI, and SQLite to provide a web-based platform for food providers, recipients, logistics coordinators, and administrators.

The project achieves its main objectives by supporting food submission, AI-based perishability analysis, recipient matching, redistribution approval, pickup tracking, and impact monitoring. Google Gemini API supports food analysis, while a rule-based fallback helps maintain functionality when the external AI service is unavailable. Overall, FoodRescue AI provides a structured approach to coordinating surplus food redistribution and connecting available food with people in need.

Future Scope:

1. Short-Term Improvements: Improve the user interface, strengthen input validation, and enhance error handling and notification management.
2. Medium-Term Improvements: Integrate live GPS tracking, configure real SMS and email notifications, and connect the existing Java Spring Boot microservices with the main application.
3. Long-Term Improvements: Introduce IoT-based temperature monitoring to support cold-chain management, develop a dedicated mobile application for delivery personnel, and use historical data to improve demand prediction and food redistribution planning.

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